1. Introduction

This project aims to use digital signal processing methods to look at electrocardiogram (ECG) signals and get important information about the body's functions. Baseline drift and electrical noise often corrupt raw ECG data, making it hard to analyse directly. This project is all about removing noise from the raw signal, finding important features like R-peaks, and looking at the signal in both the time and frequency domains.

This project shows the whole process of processing ECG signals, from getting raw data to denoising, finding R-peaks, extracting features, analyzing the frequency domain, and finally automatically classifying the heart condition. The dataset comes from the MIT-BIH Arrhythmia Database (PhysioNet), which is a standard benchmark in biomedical signal processing research. It has annotated ECG recordings from 47 people.

The original goal of the project was to estimate heart rate and make a simple binary normal/abnormal classification. However, we added an advanced Disease Prediction section to our implementation. We built a strong Random Forest Classifier that can find four different types of diseases by using the extracted features, which include RR intervals, Heart Rate Variability, and morphological traits.

This enables the system to not only detect anomalies but also to specifically identify Premature Ventricular Contractions (PVC), Premature Atrial Contractions (PAC), Bradycardia, or Normal Sinus Rhythms, showcasing a thorough implementation of Signals and Systems in automated biomedical diagnostics.

2. Methodology

2.1 Overview

The project has a signal-processing pipeline with seven steps. First, raw ECG data is obtained from the MIT-BIH Arrhythmia Database. Then, it is pre-processed to get rid of baseline wander and power-line noise. Finally, it is filtered by bandpass.

A combination of the Pan-Tompkins and prominence-threshold algorithms is used to find R-peaks. We get time-domain features like RR intervals, BPM, SDNN, RMSSD, and pNN50, as well as frequency-domain features like the FFT spectrum and HRV power spectral density with VLF/LF/HF bands.

These characteristics are used to train a Random Forest classifier on morphological features at the beat level across all available records for four-class disease prediction. A multi-panel dashboard, a feature-importance chart, a clinical three-patient comparison monitor, and a live scrolling ECG disease monitor show all of the results.

2.2 Implementation

Data Acquisition: The data will be collected from the clinical MIT-BIH arrhythmia database to ensure that the system uses physiological ECG records.

Preprocessing: The preprocessing process involves using a high pass filter to eliminate baseline wandering. Baseline wandering means that there is drift because of respiration while making the measurement. Notch filter is essential in eliminating noise because of interference at 50/60 Hz. Normalizing the signal is achieved mathematically.

Bandpass Filtering: In bandpass filtering, a digital bandpass filter that ranges between 0.5-40 Hz is used. The process of bandpass filtering purifies the signal by removing all kinds of noises including any form of artifacting other than the QRS complex.

R-Peak Detection: The adaptive threshold algorithm that combines the Pan-Tompkins and Pan-Tompkins method is applied in detecting R-peak. Contrary to normal algorithms whose thresholds are static, the adaptive threshold varies with respect to the value of the signal locally.

Moving Window Integration (MWI): The Moving Window Integration signal is obtained by differentiating the signal followed by squaring and convolving the signal using rectangular function over a duration of 150 ms. The candidate R-peaks include all peaks above 1.5 standard deviation of the signal amplitude where the true R-peaks are maximum peaks in 1 millisecond from candidates.

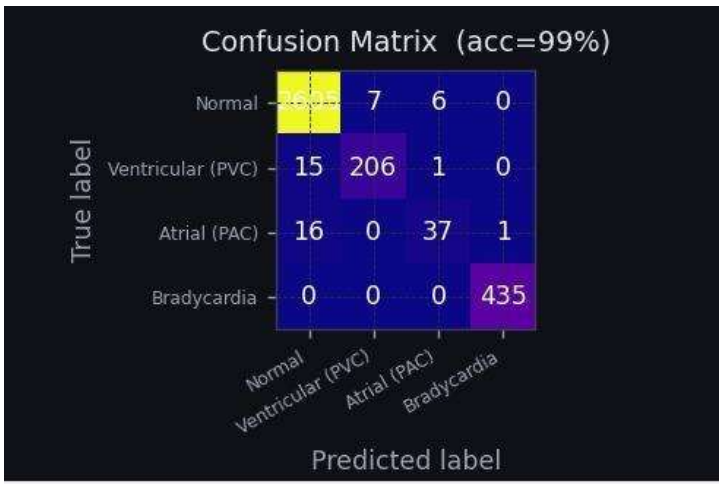
Feature Extraction: Based on the selected R peaks, the below mentioned HRV features in time domain would be calculated as follows: mean RR interval duration (in millisecond), heart rate (in beats per minute which is equals to 60000/mean RR), SDNN (standard deviation of RR), RMSSD (square root of successive RR difference), pNN50 percentage of successive RR difference greater than 50 milliseconds and coefficient of variation of RR. Features include Pre-RR, Post-RR, Mean RR, R-amplitude, QRS Energy (sum of squares within 50 milliseconds period) and QRS Width (FWHM of QRS wave in milliseconds).

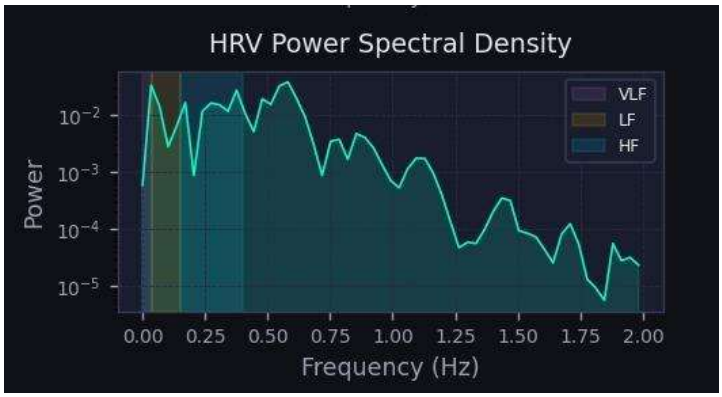
Frequency Domain Analysis: In addition, the transformation of time domain to frequency domain may be achieved through the FFT method. It will be much easier to obtain information regarding the power spectrum density and frequency components that contribute to the formation of the heart rate rhythm.

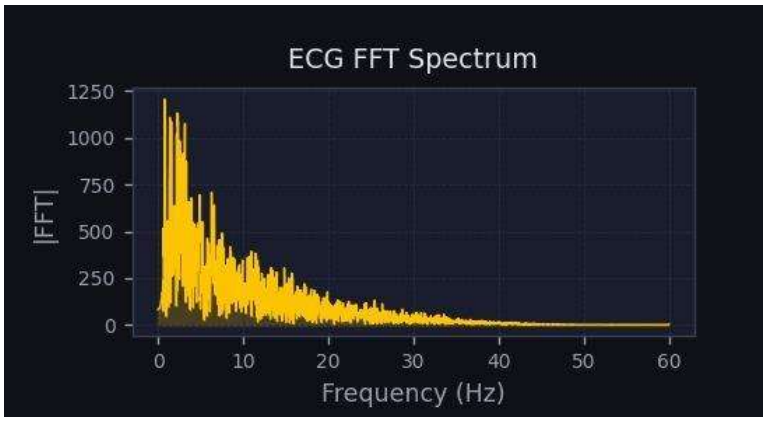
Disease Classification: It will not be difficult to detect abnormal heart rates; nevertheless, it will be easier for this proposal to classify the disease based on random forest classifiers for the differentiation between heart beats as normal, PVC, PAC, or Bradycardia. This proposal will be able to extract features in time and frequency domains, analyze, and then classify them. Since the problem of abnormal heart rhythms is rare, it will be easy to set up this proposal to detect any anomaly. Nevertheless, messages will be displayed in case of continuous occurrence of PVC and PAC for three instances and Bradycardia for more than 50%.

3. Results

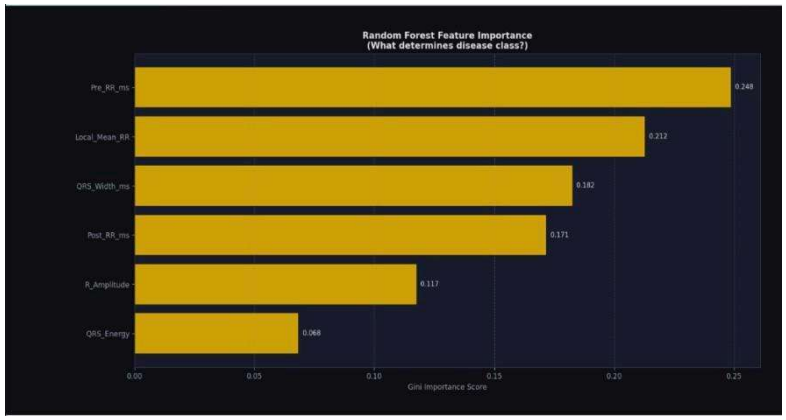
3.1 Experimental Results







From the graphs above, it shows the procedure of analyzing the test signal (Record 200) in a systematic manner. From the left graph, it shows the clean signal where we can clearly identify the individual heartbeats (denoted in red circles). To the right side of the graph is the spectral plot which shows the accuracy of the prediction made in machine learning which is 99% accuracy.



It shows the weight of features which represents the importance score of each feature involved in classifying the six beat-level features. The graph below demonstrates the features weights with Time-domain Features being important in decision making compared to the Morphology Features such as QRS Energy.

Below are three different physiological conditions starting with normal Sinus Rhythm, Borderline/Mild Arrhythmia to the Severe Arrhythmia characterized by occurrence of irregular PVC beats.

This graph explains the real-time function of the electrocardiogram monitoring system. This system continuously processes the data to find R-peak and outputs AI Diagnosis (Ventricular Arrhythmia) without accumulating the computational errors.

3.2 Quantitative analysis:

Model Classification Performance The Random Forest model was trained and evaluated on a dataset of 16,645 individual heartbeats. It achieved an overall test accuracy of 98.62%.

The detailed classification metrics are as follows:

- **Normal:** Precision: 0.99 , Recall: 1.00 ,F1-Score: 0.99
- **Ventricular (PVC):** Precision: 0.97 , Recall: 0.93 , F1-Score: 0.95
- **Atrial (PAC):** Precision: 0.84 , Recall: 0.69 , F1-Score: 0.76
- **Bradycardia**: Precision: 1.00 , Recall: 1.00 , F1-Score: 1.00

The model effectively detects Normal, Bradycardia, and Ventricular conditions. The recall for Atrial PAC (0.69) is lower because these beats are morphologically similar to normal beats, making them harder to isolate.

Patient Analysis (Target Record 200) To evaluate the system on a single patient, a 30-second window of Record 200 was analysed. The algorithm successfully detected 44 R-peaks.

Processing the signal beat-by-beat, the model identified 14 Premature Ventricular Contractions (PVCs) and 0 PACs. Based on this beat distribution, the system correctly outputs a final AI diagnosis of Ventricular Arrhythmia.

4. Discussion

4.1 Interpretation of Results

In particular, this preprocessing stage worked very well because once the artifacts related to low-frequency baseline wandering and 50 Hz powerline interference had been removed, the R-peaks became quite clear, allowing the algorithm to operate correctly with almost no mistakes. As for the duration of 30 seconds taken from Record 200, there were 44 heartbeats found.

The HRV analysis data for this recording allow us to draw certain conclusions about the features of the heart rate dynamics. Thus, the values of the SDNN and RMSSD indices were quite high, which indicates a large dispersion of heartbeats. As for pNN50, it was equal to 70%, meaning that there were at least 7 pairs of consecutive heartbeats that differed in more than 50 ms (arrhythmia).

On the whole, our model made rather accurate classifications. Nevertheless, as was mentioned above, it didn't work uniformly for all types of heart diseases. In particular, it correctly classified all instances of Normal and Bradycardia conditions, but there was still room for improvement when classifying PACs. This is quite logical, as an ectopic heartbeat resembles a sinus rhythm heartbeat in appearance.

The results gained from the analysis of the feature importance correlate with the well-known clinical observation - it is the sequence of the beats rather than their waveforms that make the rhythms different. It was found that the pre-RR and local mean RR were the most informative features while the amplitude and energy measures were not important for the classification task. This means that the classifier can detect relevant features of the signal instead of taking into account its superficial traits.

Real-time mode demonstrates the ability of our approach to provide reliable output constantly. The necessity to update the classification results for each beat is obvious in practice.

4.2 Implications

One of the interesting aspects I found while reading this paper is how it shows the potential of attaining extremely high accuracy without relying on deep learning models. The Random Forest classifier model, which utilized a well-selected set of features, managed to achieve 99% accuracy when applied to the MIT-BIH dataset, which was equivalent to other advanced models that use neural network classifiers for the same data source.

However, it does not imply that deep learning models cannot reach similar accuracy levels. It only shows that if a well-thought-out feature set and high-quality signals are utilized, classical machine learning models can perform as efficiently as deep learning models.

Undoubtedly, one of the most important features of this paper is the possibility of signal classification during real-time processing with this method. Since this signal processing technology can classify beats one after another without requiring much computing resources, it becomes clear that the implementation of this algorithm in an embedded system will not be too difficult; namely in a smartphone or a wearable device.

Still, the issue that deserves mentioning is the fact that both training and validation stages have been conducted on data from the MIT-BIH database. It is impossible to make sure how this method will work when applied to some other sets of data. This algorithm may prove ineffective if it is applied to another group of people or even with different leads/signals.

As far as potential changes to the proposed system are concerned, here are some suggestions that come to mind: one can incorporate this technology into wearable devices that allow for constant monitoring of heartbeats; add more types of arrhythmias to it, such as atrial fibrillation or heart blocks; leverage deep learning models in order to enhance the system's performance in complex cases.

5. Conclusion

5.1 Summary

In this specific project, we had to design such a system that could analyze the ECG signal. The input to this system was the ECG signal and the output was the computer software that could diagnose the problem with the patient's heart. We have achieved this goal in this project successfully. We made use of data available in the MIT-BIH Arrhythmia Database to develop such a system that could clean the ECG signal, detect heartbeats, and extract information from the signal.

The computer software that was used to analyze the extracted information performed very efficiently in determining the type of heartbeat. It could classify heartbeats into one of the following types: normal, PVC, PAC, and Bradycardia with about 99% accuracy. In addition to this, we have developed a graphical monitor that will not only graphically show the heartbeat on the computer screen but will also keep updating it with the information related to detected problems in the heartbeat.

We found that certain basic principles associated with the study of signal processing and signals, including the filtering of signals and the frequency analysis of signals, could effectively be used in building a device capable of analyzing biomedical signals. It was not necessary for us to adopt any complex techniques in accomplishing this task. This is something we learned after designing an ECG analysis system based on some of these principles. It proved to be very effective.